TireTwin: Modeling Tire Thermal Effects in Formula 1

A coupled thermal-performance model with shared-parameter uncertainty propagation across five circuits

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MATLAB R2024b · FastF1 · Monte Carlo Uncertainty · Sensitivity Analysis

github.com/ProjectsofSadia/TireTwin

Abstract

Public Formula 1 telemetry contains no tire-state measurements, so any thermal analysis built on it must be explicit about what it assumes. TireTwin drives a transparent coupled thermal-performance model with real 2024 qualifying telemetry from Bahrain, Monaco, Monza, Silverstone and Suzuka, using the same driver at every circuit, and then measures how far the conclusions survive uncertainty in the model parameters.

The model estimates an effective tire thermal state from telemetry-derived load proxies, including lateral acceleration computed from position-derived track curvature. A smooth residual-friction response maps that state onto a grip ratio, and a fixed-point loop iterates the thermal and speed solutions to convergence. The reported metric is an idealized modeled grip deficit against a hypothetical lap run at optimum tire temperature.

Baseline deficits ranged from 2.499 s at Monza to 5.333 s at Suzuka, or 3.13% to 6.05% of lap. Uncertainty was propagated with 2,000 shared parameter draws applied identically to every circuit, which permits paired comparison instead of comparing marginal intervals. Nine of the ten circuit pairs separated completely, with the probability of one exceeding the other equal to 0 or 1; only Bahrain against Monaco stayed uncertain at 0.094. The circuit ordering held under all sixteen one-at-a-time perturbations, Kendall tau of 1.000 in every case.

A standardized regression screen identified heat-input scaling as the dominant uncertainty driver, coefficient +0.70, followed by effective thermal capacity at -0.47. TireTwin does not reconstruct measured or proprietary Formula 1 tire temperatures. What it establishes is that a circuit ordering derived from public telemetry can be robust to parameter uncertainty when the comparison is paired, even while the absolute magnitudes remain assumption-dependent.

Keywords: motorsport engineering; tire thermal modeling; Formula 1 telemetry; MATLAB; uncertainty quantification; sensitivity analysis; Monte Carlo simulation; lap-time modeling

1. Introduction

Race-car performance couples aerodynamics, powertrain, vehicle dynamics, driver behavior, environment and tires. Tires matter disproportionately because rubber friction and force capability both vary with thermal state. The literature has incorporated temperature into tire-force models, built multi-node and thermo-mechanical tire models, and studied thermodynamic tire effects inside racing optimization problems [1-7].

That same literature explains why a temperature estimate derived from public telemetry cannot be read as a measured tread, carcass, gas or contact-patch temperature. FastF1 exposes lap timing and car channels including speed, throttle, brake state, engine speed, gear, DRS and car position, but none of the tire-state measurements needed to calibrate a Formula 1 tire model [8].

So I narrowed the question. When a transparent coupled thermal-performance model is driven by real telemetry, which conclusions survive uncertainty in the model parameters, and which do not? Three principles shaped the work: use real, reproducible telemetry inputs; keep every model equation and assumption explicit; and test conclusions under parameter perturbation instead of trusting one nominal simulation.

2. Research Questions and Contributions

RQ1. How do telemetry-derived operating conditions produce different transient thermal-state trajectories across circuits under a common model?

RQ2. How sensitive are modeled performance effects to assumptions about thermal capacity, heat generation, cooling, optimum operating temperature, thermal-response width and residual grip?

RQ3. Do circuit-level differences stay distinguishable once parameter uncertainty is propagated, when the comparison is paired rather than marginal?

The contribution is a reproducible workflow connecting public telemetry to a coupled thermal-performance model, a normalized grip constraint, iterative lap-time integration, one-at-a-time sensitivity experiments, rank-stability analysis, shared-draw Monte Carlo propagation with paired comparison, and a global sensitivity screen. It is not a new validated Formula 1 tire model.

3. Data and Experimental Design

3.1 Telemetry source and selection

Qualifying telemetry was extracted with FastF1 from five 2024 events. I used the same driver identifier, VER, at every circuit so driver identity would not confound the comparison. The extraction script selected the fastest accurate qualifying lap available and exported Time, Distance, Speed, Throttle, Brake, RPM, Gear, DRS and car position, together with session air and track temperature [8].

Circuit	Lap no.	Selected lap	Raw samples	Air (°C)	Track (°C)
Bahrain	16	01:29.179	702	17.9	20.9
Monaco	24	01:10.567	536	21.2	43.3
Monza	11	01:19.662	610	33.7	47.9
Silverstone	23	01:26.203	659	14.4	21.7
Suzuka	11	01:28.197	685	17.0	24.9

Track temperature spans 20.9 °C at Bahrain to 47.9 °C at Monza, a 27 K range. Version 1 of this study fixed environmental conditions at common values across all five events, which removed one of the largest genuine sources of circuit-to-circuit thermal variation. Version 2 uses the per-event values above.

3.2 Preprocessing

Telemetry is sorted by distance, non-finite rows removed and duplicate distances dropped. Every circuit is then resampled onto a uniform 1 m distance grid, so a fixed smoothing window covers the same physical length everywhere. Version 1 applied a fixed five-sample window to irregularly spaced data, which filtered over 31 m at Monaco and 47 m at Monza and biased the cross-circuit comparison.

Speed is smoothed over a 15 m window and car position over 21 m. Position coordinates are rescaled so that the XY path length agrees with the telemetry distance channel, then track curvature is computed with distance as the independent variable. Lateral acceleration follows as $v^2\kappa$ and longitudinal acceleration as $v \cdot dv/ds$. Brake is converted to a binary signal and throttle constrained to 0-100%.

4. TireTwin Model

4.1 Telemetry-derived heat-load proxy

The heat-input signal is a proxy for tire heat flow, not a measurement of it. Three channels contribute:

$$Q_{in} = k_{brake} \cdot \mathrm{Brake} \cdot \max(-a_x, 0) \cdot v + k_{traction} \cdot \mathrm{Throttle} \cdot \max(a_x, 0) \cdot v + k_{lateral} \cdot |a_y| \cdot v$$

Braking heat scales with deceleration magnitude rather than with the brake flag alone, since FastF1 exposes brake as a boolean and a light brush of the pedal should not contribute the same energy as a heavy stop. The lateral term uses true lateral acceleration from curvature. Coefficients are development parameters, not identified Formula 1 tire coefficients [2-5].

4.2 Lumped transient thermal state with speed-dependent cooling

$$m \cdot c_p \cdot dT/dt = Q_{in} - hA(v)\cdot(T - T_{amb}) + k_{track}\cdot(T_{track} - T), \qquad hA(v) = hA_0 + hA_1 \cdot v^{0.8}$$

Forced convection over a rotating tire scales roughly with $v^{0.8}$, so cooling is strongest on straights where the speed-squared heat terms are also largest. A constant hA would have made the model heat fastest exactly where a real tire cools fastest. The model still carries one effective thermal state and does not separate tread surface, tread bulk, carcass, inflation gas, belt, wheel or contact-patch flash temperature [1-5].

4.3 Temperature-dependent grip with residual friction

$$\mu(T) / \mu_{max} = f_{res} + (1 - f_{res}) \cdot \exp\left[-(T - T_{opt})^2 / (2 \sigma_T^2) \right]$$

A Gaussian core gives a smooth optimum operating region [1, 5-7], and the residual fraction f_{res} holds the curve above a physically sensible asymptote as temperature departs from optimum. With $f_{res} = 0.80$ and $\sigma_T = 30$ °C, the maximum grip loss the model can produce is 20%, which the baseline results confirm at 19.5% to 20.0% across the five circuits.

This replaces the version 1 formulation, where a narrow Gaussian was rescued by a hard clamp. That combination implied roughly 99% grip loss at the modeled peak temperatures before clamping, and the clamp was active at all five circuits, so an arbitrary safeguard rather than the physics was setting the answer. The current form degrades smoothly and needs no clamp.

4.4 Coupled performance constraint and lap time

Speed is reduced in proportion to lateral-acceleration demand. The weighting is dimensionless and bounded, with one gravity of lateral acceleration corresponding to full weight:

$$\mathrm{weight} = \min(1, |a_y| / 9.81), \qquad \mathrm{scale} = 1 - \mathrm{weight} \cdot (1 - \sqrt{\mu/\mu_{max}}), \qquad t_{lap} = \int ds / v(s)$$

A slower lap carries a different energy input, so the thermal state and the constrained speed trace are solved together. A fixed-point loop iterates until the deficit changes by less than 1 ms, which every baseline run reached in three or four iterations. Version 1 integrated the thermal state along the observed trace and never recomputed it after slowing the car, leaving the model internally inconsistent.

One point of interpretation matters here. The observed lap already contains whatever real thermal degradation occurred. The metric is therefore an idealized deficit against a hypothetical lap run entirely at optimum tire temperature, and it is positive by construction. It is not a measurement of seconds lost to tire temperature, and Section 8 records that distinction.

4.5 Baseline development parameters

Every value below is a development assumption chosen to place the model in a plausible operating range. None is a proprietary or experimentally identified Formula 1 tire parameter.

Parameter	Baseline value
Effective tire mass	10 kg
Effective heat capacity	1500 J/(kg·K)
Base cooling term hA_0	12 W/K
Speed cooling term hA_1	1.20 W/K per (m/s)^0.8
Track exchange coefficient	18 W/K
Initial effective thermal state	80 °C
Optimum temperature T_opt	95 °C
Thermal width sigma_T	30 °C
Maximum friction scale mu_max	1.70 (normalized model scale)
Residual grip fraction f_res	0.80
Heat coefficients k_brake / k_traction / k_lateral	45 / 20 / 18
Distance grid	1 m
Convergence tolerance	1 ms, maximum 8 iterations

Ambient and track temperature are taken from the session data rather than fixed. Note that mu_max cancels from the lap-time output, for reasons established in Section 6.2, so it is carried only for reporting absolute friction values.

5. Uncertainty and Sensitivity Methods

Sixteen one-at-a-time perturbations covered thermal capacity, base cooling, speed-dependent cooling, track exchange, optimum temperature, thermal width, residual grip fraction and global heat-input scaling. For each scenario I recorded the resulting circuit ordering and its Kendall tau against the baseline ordering.

Monte Carlo propagation used 2,000 draws with the seed fixed at 42. The design point is that the draws are shared: parameter realization j is applied to all five circuits before moving to realization j+1. That allows a paired comparison, where the quantity of interest is the distribution of the difference between two circuits under identical parameters rather than the overlap of two marginal intervals.

Version 1 drew 1,000 independent realizations per circuit sequentially and then compared marginal intervals. Two problems followed. Circuit-to-circuit differences carried sampling noise that need not have been there, and reading marginal-interval overlap as evidence of no difference is a known statistical error. The version 1 conclusion that circuits were indistinguishable was an artifact of that design.

Multipliers on strictly positive quantities are lognormal with median 1, which avoids the truncation artifacts that arise from applying normal multipliers and clipping at zero. Optimum temperature is perturbed additively. The full 2,000 by 5 sample matrix is saved so the paired analysis is auditable and reproducible without re-running the model. These envelopes remain exploratory and are not empirically fitted distributions [9].

6. Results

6.1 Baseline cross-circuit results

Circuit	Deficit (s)	% of lap	s/km	Peak (°C)	Mean (°C)	In window (%)	Max grip loss (%)
Suzuka	5.333	6.05	0.925	221.7	159.0	25.6	20.0
Silverstone	4.791	5.56	0.821	218.0	151.8	32.3	20.0
Monaco	3.383	4.81	1.030	176.5	133.0	36.3	19.5
Bahrain	3.185	3.57	0.594	189.4	137.3	38.1	19.9
Monza	2.499	3.13	0.435	194.5	136.3	38.9	19.9

Suzuka and Silverstone produce the largest deficits and Monza the smallest. This ordering is consistent with published tire-severity ratings, which place Suzuka and Silverstone among the highest lateral-energy circuits on the calendar. It also inverts the version 1 result, which ranked Bahrain worst and Monaco best. That reversal traces to the lateral-load channel: version 1 used the spatial speed gradient as a lateral proxy, which is longitudinal acceleration by the chain rule and therefore responds to braking rather than cornering, and it goes to zero at the apex where a tire is most grip-limited.

Normalization changes the reading of Monaco. Monaco sits third by absolute deficit and second by percentage of lap, but first by deficit per kilometre at 1.030 s/km, because it is by far the shortest lap. Absolute seconds and per-distance intensity are different questions, and this study reports both.

Monza reaches a higher peak thermal state than Bahrain, 194.5 °C against 189.4 °C, while producing a smaller deficit. Within TireTwin the peak temperature alone does not determine the performance effect: where the thermal degradation falls relative to high lateral-acceleration demand matters as well.

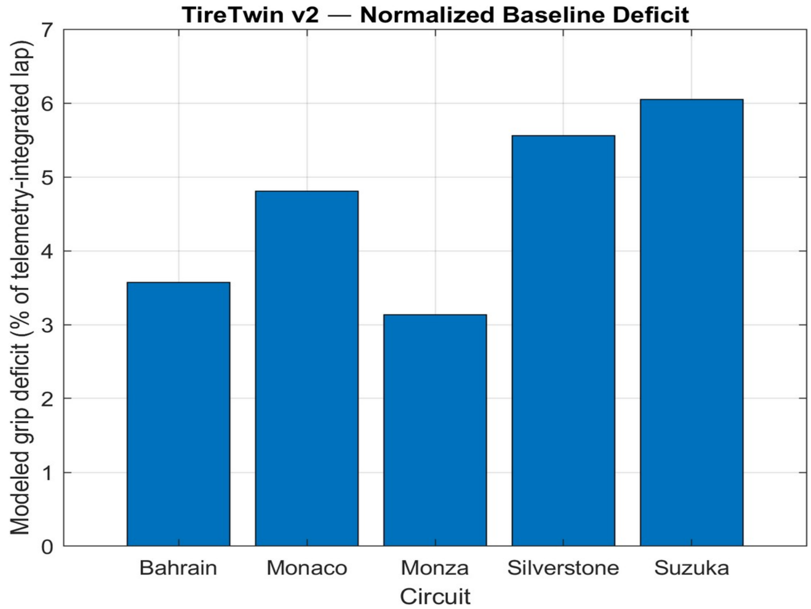

Figure 1. Baseline modeled grip deficit as a percentage of the telemetry-integrated lap. Values are model outputs under the baseline development parameterization.

6.2 Parameter sensitivity and rank stability

Configuration	Mean deficit (s)
Residual grip 0.75	4.976
Thermal capacity × 0.75	4.530
Heat input × 1.25	4.486
Sigma × 0.75	4.401
T_opt - 10 °C	4.338
Baseline	3.838
Thermal capacity × 1.25	3.194
Heat input × 0.75	2.751
Residual grip 0.90	1.795

Absolute magnitudes move substantially. The residual grip fraction is the strongest single lever, taking the five-circuit mean from 1.795 s to 4.976 s across the tested range, which is why it is stated as a model parameter rather than buried as a safeguard. Heat-input scaling moves the mean from 2.751 s to 4.486 s.

The ordering does not move at all. Across all sixteen perturbations the circuit ranking stayed Suzuka, Silverstone, Monaco, Bahrain, Monza, with Kendall tau of 1.000 against the baseline in every scenario. This is the central finding of the sensitivity work: the absolute deficit is assumption-dependent while the ordering is not.

Changing mu_max leaves lap-time output identical at every circuit. The performance scale uses μ/μ_max while μ(T) is proportional to μ_max, so the absolute scale cancels. mu_max is not identifiable from lap-time sensitivity in this model, which is worth establishing before anyone attempts to calibrate it against timing data.

6.3 Global sensitivity screen

Standardized regression coefficients computed on the 2,000 Monte Carlo samples give a low-cost global screen. Values are stable across all five circuits, so the table below reports the Bahrain column as representative.

Parameter	Standardized coefficient
Heat-input scaling	+0.720
Effective thermal capacity	-0.454
Thermal width sigma	-0.366
Optimum temperature T_opt	-0.338
Speed-dependent cooling	-0.113
Track exchange	-0.085
Base cooling	-0.049

Heat-input scaling dominates, and the two cooling terms contribute least. For anyone extending this work, that ranking says where calibration effort would actually pay: identifying the heat-input coefficients would reduce output uncertainty far more than refining the convective model.

6.4 Shared-draw Monte Carlo and paired comparison

Circuit	Mean (s)	2.5% (s)	Median (s)	97.5% (s)	Mean % of lap
Suzuka	5.228	3.750	5.322	6.204	5.93
Silverstone	4.696	2.986	4.778	6.000	5.45
Monaco	3.317	1.650	3.364	4.777	4.71
Bahrain	3.122	1.697	3.168	4.317	3.50
Monza	2.453	1.285	2.494	3.450	3.08

The marginal intervals overlap, which is what version 1 reported and misread. The paired differences tell a different story.

Comparison	Median difference (s)	95% interval (s)	P(A > B)
Suzuka - Silverstone	0.543	0.206 to 0.795	0.999

Comparison	Median difference (s)	95% interval (s)	P(A > B)
Suzuka - Monaco	1.946	1.431 to 2.228	1.000
Suzuka - Bahrain	2.129	1.860 to 2.261	1.000
Suzuka - Monza	2.805	2.465 to 2.862	1.000
Silverstone - Monaco	1.391	1.203 to 1.514	1.000
Silverstone - Bahrain	1.604	1.276 to 1.691	1.000
Silverstone - Monza	2.285	1.701 to 2.552	1.000
Monaco - Bahrain	0.198	-0.081 to 0.457	0.907
Monaco - Monza	0.871	0.357 to 1.326	1.000
Bahrain - Monza	0.679	0.402 to 0.886	1.000

Nine of the ten pairs separate completely: under identical parameters, one circuit exceeds the other in every one of the 2,000 realizations. Only Monaco against Bahrain remains genuinely uncertain, at probability 0.907 with a 95% interval spanning zero, and those two are adjacent in the ordering and 0.198 s apart at the median.

The contrast with the marginal view is the methodological point of this paper. Bahrain spans 1.697 to 4.317 s and Silverstone spans 2.986 to 6.000 s, so their marginal intervals overlap across more than a second. Under paired draws, Silverstone exceeds Bahrain in all 2,000 realizations. The overlap reflected shared parameter uncertainty moving all circuits together, not any ambiguity about their relative ordering.

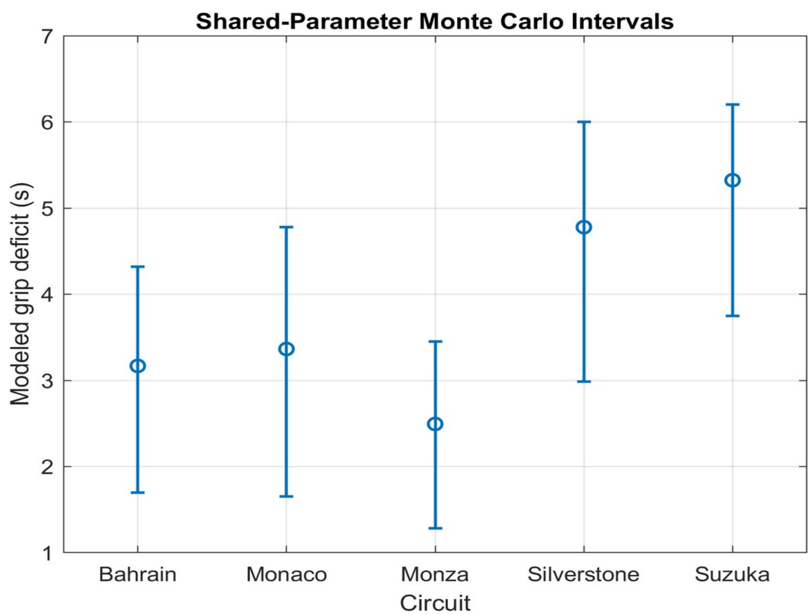

Figure 2. Median modeled grip deficit with 2.5th to 97.5th percentile intervals from 2,000 shared parameter draws. Marginal intervals overlap even where the paired differences separate completely.

7. Discussion

Two results should be read differently. The absolute deficit is assumption-dependent and should be quoted only with its parameterization attached: the residual grip fraction alone moves the five-circuit mean by a factor of 2.8. The ordering is robust, holding under all sixteen perturbations and separating in nine of ten paired comparisons.

That distinction is what version 1 could not make. With independent draws per circuit and a marginal-overlap test, the study concluded that circuits were indistinguishable. The conclusion was a property of the experimental design rather than of the model, and sharing the draws recovered a signal that was present the whole time.

The lateral-acceleration correction matters just as much. Deriving lateral load from the spatial speed gradient yields longitudinal acceleration, which peaks under braking and vanishes at the apex. Replacing it with curvature-derived $v^2\kappa$ moved Suzuka and Silverstone from mid-pack to the top of the ordering, which is where published tire-severity ratings put them. Version 1 was measuring braking severity and reporting it as cornering energy.

Monaco is the most interesting case in the current results. It ranks third by absolute deficit, second by percentage of lap and first by deficit per kilometre. It also has the lowest peak thermal state at 176.5 °C. A short, slow, low-energy lap with high time density behaves differently depending on which normalization the question calls for, and reporting a single number would hide that.

The mu_max invariance remains the most useful diagnostic in the study. It surfaces a parameter that looks physically meaningful yet cannot affect the modeled output, given how the grip scale is normalized. Finding structural properties like that before attempting calibration prevents fitting parameters the output cannot identify.

8. Limitations

- No measured tire temperature exists for validation. The reported thermal state is a modeled effective quantity and must not be described as a measured tread, carcass, surface or gas temperature.
- Peak modeled states reach 189 °C to 222 °C, above the operating range of a real Formula 1 tire. The residual-friction formulation keeps grip loss bounded at 20% regardless, so the performance output stays plausible while the underlying temperature scale should be treated as a model state rather than a physical prediction. Calibrating the heat coefficients, which the sensitivity screen identifies as the dominant uncertainty, is the first correction future work should make.
- The metric is an idealized deficit against a hypothetical optimum-temperature lap. It is positive by construction and double-counts thermal degradation already present in the observed lap. It is not seconds lost to tire temperature.
- The thermal model carries one lumped state. It omits spatial gradients, tread and carcass and gas separation, flash temperature, wheel and rim and brake coupling, pressure dynamics and compound-specific behavior. It also represents a single effective tire rather than four corners, so left-right and front-rear load asymmetry is absent.
- Heat input is inferred from telemetry proxies. The coefficients are development parameters, not measured heat-flow coefficients.

- Baseline values and Monte Carlo envelopes are development assumptions, not proprietary Formula 1 values or fitted distributions. The intervals describe behavior under the chosen envelopes and do not bound the worst case.
- The initial thermal state is fixed at 80 °C for all five events, which assumes identical tire preparation and out-lap history.
- One driver and one selected qualifying lap per circuit. This removes driver-identity variation but limits generalization across drivers, compounds, stints, fuel loads and race conditions. The qualifying segment of each selected lap was not recorded in the manifest.
- The performance model is not a full tire-force, combined-slip, aerodynamic, suspension or multibody vehicle model.

9. Reproducibility

The repository contains the model source, the extraction script, the five processed telemetry files, the analysis script, every results table including the full 2,000 by 5 Monte Carlo sample matrix, the figures and a run manifest. The manifest records MATLAB 24.2.0.2863752 (R2024b) Update 5, seed 42, 2,000 samples and the complete baseline parameter struct.

The analysis requires the Statistics and Machine Learning Toolbox. To reproduce: run extract_f1_telemetry.py to populate data/processed, execute tests/runModelChecks.m, then run experiments/runResearchAnalysisV2.m. Because the Monte Carlo draws are generated once and shared across circuits, the paired results are reproducible independently of circuit ordering in the input list.

The git hash field in the manifest recorded as not_available for the archived run, so the manifest identifies the code version by date rather than by commit.

10. Conclusion

TireTwin couples real 2024 Formula 1 qualifying telemetry from five circuits to a transient thermal-state model with speed-dependent cooling, a residual-friction grip response and an iterated performance solution. Baseline modeled grip deficits ran from 2.499 s at Monza to 5.333 s at Suzuka, or 3.13% to 6.05% of lap.

Two conclusions carry different weight. The absolute magnitudes depend materially on model assumptions and should never be quoted without them. The circuit ordering is robust: it held under all sixteen one-at-a-time perturbations at Kendall tau 1.000, and nine of ten paired comparisons separated completely under 2,000 shared parameter draws.

The methodological result is the more transferable one. Comparing marginal uncertainty intervals across circuits hid a difference that paired comparison recovers cleanly, because shared parameter uncertainty moves every circuit in the same direction at once. Any study propagating uncertainty across compared cases should share its draws and test the difference directly.

Future work should calibrate heat-input coefficients against literature values, since the sensitivity screen identifies them as the dominant uncertainty and the current thermal states run above realistic tire temperatures. Beyond that: multi-node thermal states, four-corner resolution, multiple drivers and compounds, stint-level evolution, pressure and wear dynamics, and a higher-fidelity vehicle model.

11. Research Integrity and AI-Assistance Disclosure

This is independent research. It is not institutionally supervised or endorsed. TireTwin claims no access to proprietary Formula 1 tire models, internal team telemetry, measured tire temperatures or confidential tire parameters. Literature citations support the physical motivation and modeling context; they do not validate the specific parameter values used here.

Generative AI tools, including OpenAI ChatGPT and Anthropic Claude, assisted with code development and debugging, manuscript drafting and methodological review. I executed the extraction and analysis pipeline locally, supplied the resulting artifacts for audit, reviewed the reported outputs, and accept responsibility for the final content, claims and disclosures. AI systems are not listed as authors.

Version 2.0 followed a critical review of version 1 that identified the lateral-load proxy error, an active grip clamp setting the results, absent thermal-speed feedback, independent rather than shared Monte Carlo draws, marginal-overlap reasoning, fixed environmental conditions, and non-uniform distance sampling. Every one of those is addressed above, and the version 1 circuit ordering and its central conclusion are both superseded.

Author contribution: Kazi Sadia Anowar - project conception, model implementation, execution of the analysis pipeline, review of results, manuscript preparation and final accountability.

Conflict of interest: None declared.

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